

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET F — RAPID FIRE SPEED TEST (30 MINS)

CHAPTER 04: DESCRIBING MOTION & KINEMATICS

TIME ALLOWED: 30 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-F-04

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET F - RAPID FIRE SPEED TEST):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **SPEED CHALLENGE INSTRUCTIONS:** Answer promptly without looking at notes. Test your instant recall of formulas, scientific definitions, and core NCERT concepts!

Q.1 The shortest straight-line distance between the initial and final positions of a moving body is its:

- | | |
|--------------|-----------------------|
| (A) Distance | (B) Displacement |
| (C) Speed | (D) Trajectory length |

Q.2 What is the SI unit of acceleration?

- | | |
|-------------------|-------------------------------|
| (A) m/s | (B) m/s^2 |
| (C) km/h | (D) $\text{m} \cdot \text{s}$ |

Q.3 What does the slope of a distance-time (s - t) graph represent?

- | | |
|------------------|-----------|
| (A) Acceleration | (B) Speed |
| (C) Displacement | (D) Force |

Q.4 What does the slope of a velocity-time (v - t) graph represent?

- | | |
|------------------|------------------|
| (A) Distance | (B) Acceleration |
| (C) Displacement | (D) Momentum |

Q.5 What does the area enclosed beneath a velocity-time graph represent?

- | | |
|-------------------|-------------------------|
| (A) Acceleration | (B) Total displacement |
| (C) Average speed | (D) Instantaneous power |

Q.6 A car accelerates uniformly from rest (0 m/s) to 20 m/s in 5 s . Its acceleration is:

- | | |
|-----------------------|-------------------------|
| (A) 2 m/s^2 | (B) 4 m/s^2 |
| (C) 5 m/s^2 | (D) 100 m/s^2 |

Q.7 Convert a speed of 72 km/h into meters per second (m/s):

- | | |
|----------------------|----------------------|
| (A) 15 m/s | (B) 20 m/s |
| (C) 25 m/s | (D) 36 m/s |

Q.8 An athlete completes one circular round of track of radius R in 40 s . His displacement after 40 s is:

- (A) $2\pi R$ (B) $2R$
(C) Zero (D) πR

Q.9 An odometer installed in a motorcycle measures:

- (A) Instantaneous speed (B) Average velocity
(C) Cumulative distance travelled (D) Engine acceleration

Q.10 Which kinematic equation correctly relates final velocity, initial velocity, acceleration, and distance?

- (A) $v = u + at$ (B) $s = ut + \frac{1}{2}at^2$
(C) $v^2 - u^2 = 2as$ (D) $v^2 + u^2 = 2as$

Q.11 A ball is thrown vertically upward with speed 19.6 m/s . How long does it take to reach maximum height ($g = 9.8\text{ m/s}^2$)?

- (A) 1.0 s (B) 2.0 s
(C) 3.0 s (D) 4.0 s

Q.12 A horizontal distance-time graph line parallel to the time axis indicates that the object is:

- (A) Moving with uniform speed (B) At rest (stationary)
(C) Accelerating rapidly (D) Moving backwards

Q.13 In uniform circular motion, the direction of linear velocity at any given point is:

- (A) Radially inward towards the center (B) Radially outward
(C) Tangential to the circular path (D) Zero

Q.14 Negative acceleration is commonly known as:

- (A) Retardation or Deceleration (B) Terminal velocity
(C) Impulse (D) Inertia

Q.15 Can average velocity be zero while average speed is positive?

- (A) Yes, if the body returns to its starting point (B) No, average velocity is always equal to average speed
(C) No, velocity can never be zero (D) Only in vacuum

Q.16 What is the acceleration of an object moving with a constant velocity of 50 m/s ?

- (A) 50 m/s^2 (B) 0 m/s^2
(C) 9.8 m/s^2 (D) 25 m/s^2

Q.17 A body dropped freely from rest falls for 3 s . What distance does it fall ($g = 9.8\text{ m/s}^2$)?

- (A) 29.4 m (B) 44.1 m
(C) 88.2 m (D) 14.7 m

Q.18 Which of the following is a vector quantity?

- (A) Distance (B) Speed
(C) Velocity (D) Time

Q.19 The motion of a freely swinging pendulum bob is an example of:

- (A) Uniform motion (B) Non-uniform accelerated motion
(C) Circular motion with constant velocity (D) Zero acceleration

Q.20 What is the speed of an object that travels 100 meters in 4 seconds?

- (A) 20 m/s (B) 25 m/s
(C) 40 m/s (D) 400 m/s

Q.21 If a car travels the first half of distance at speed v_1 and the second half at speed v_2 , its average speed is:

- (A) $\frac{v_1 + v_2}{2}$ (B) $\frac{2v_1v_2}{v_1 + v_2}$
(C) $\sqrt{v_1v_2}$ (D) $\frac{v_1v_2}{2}$

Q.22 What is the angle between velocity and centripetal acceleration in uniform circular motion?

- (A) 0° (B) 45°
(C) 90° (D) 180°

Q.23 A train slows down from 30 m/s to rest in 15 s . Its retardation is:

- (A) 1 m/s^2 (B) 2 m/s^2
(C) 0.5 m/s^2 (D) 4.5 m/s^2

Q.24 If the velocity of a particle is given by $v = 4t$, its motion is:

- (A) Uniform motion (B) Motion with uniform acceleration
(C) Motion with non-uniform acceleration (D) Stationary

Q.25 Which graph represents uniform motion?

- (A) A distance-time graph that is a straight line sloping upward from the origin
(B) A distance-time graph that is curved like a parabola
(C) A velocity-time graph with a steep positive slope
(D) A distance-time graph that curves downward

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID EXPLANATIONS (SET F)

Q. No.	Correct	Scientific Rationale & Formula Summary
Q.1	(B)	Displacement is the vector distance representing the straight-line directed path from start to finish.
Q.2	(B)	Acceleration is rate of change of velocity: $\frac{[\text{m/s}]}{[\text{s}]} = \text{m/s}^2$.
Q.3	(B)	Slope $= \frac{\Delta s}{\Delta t}$, which is the definition of speed.
Q.4	(B)	Slope of v - t graph $= \frac{\Delta v}{\Delta t}$, which gives the acceleration.
Q.5	(B)	Area $= v \times t$, which dimensional analysis and integration show equals displacement.
Q.6	(B)	$a = \frac{v - u}{t} = \frac{20 - 0}{5} = 4 \text{ m/s}^2$.
Q.7	(B)	$72 \times \frac{5}{18} = 4 \times 5 = 20 \text{ m/s}$.
Q.8	(C)	After one full round, the athlete returns to the exact starting position; net displacement is zero.
Q.9	(C)	An odometer measures distance covered in km; a speedometer measures instantaneous speed.
Q.10	(C)	The third kinematic equation of uniformly accelerated motion is $v^2 - u^2 = 2as$ (or $v^2 = u^2 + 2as$).
Q.11	(B)	At the peak, $v = 0$. Using $v = u - gt \Rightarrow 0 = 19.6 - 9.8 t \Rightarrow t = 19.6 / 9.8 = 2.0 \text{ s}$.
Q.12	(B)	A zero slope (horizontal line) on an s - t graph means position is not changing with time; the object is at rest.
Q.13	(C)	The instantaneous velocity vector is always directed tangentially to the circular trajectory.
Q.14	(A)	When acceleration opposes the direction of velocity, speed decreases, termed retardation or deceleration.
Q.15	(A)	If displacement is zero (round trip), $\text{average velocity} = 0$, but total distance > 0 gives a positive average speed.
Q.16	(B)	Velocity is not changing; $\Delta v = 0 \Rightarrow a = 0$.
Q.17	(B)	$s = \frac{1}{2}gt^2 = 0.5 \times 9.8 \times (3)^2 = 4.9 \times 9 = 44.1 \text{ m}$.
Q.18	(C)	Velocity specifies both magnitude (speed) and direction of motion, making it a vector.
Q.19	(B)	The bob speeds up toward the center and slows down toward the extremes, undergoing non-uniform periodic acceleration.
Q.20	(B)	$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{100 \text{ m}}{4 \text{ s}} = 25 \text{ m/s}$.
Q.21	(B)	Average speed over equal distance segments is the harmonic mean: $v_{\text{avg}} = \frac{2 v_1 v_2}{v_1 + v_2}$.
Q.22	(C)	Velocity is tangential while centripetal acceleration is directed radially inward; they are mutually perpendicular (90°).
Q.23	(B)	$a = \frac{0 - 30}{15} = -2 \text{ m/s}^2$. Retardation $= +2 \text{ m/s}^2$.
Q.24	(B)	Acceleration is constant: $a = \frac{dv}{dt} = 4 \text{ m/s}^2$, representing uniform acceleration.
Q.25	(A)	A straight-line s - t graph passing through the origin indicates constant slope, meaning uniform speed.